PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: SellerMate

Version: 1.0

1. Project Overview

- **Problem Statement:** Social commerce sellers today face increasing operational complexity as their businesses scale across chat-based platforms such as Instagram, WhatsApp, and TikTok. Unlike traditional e-commerce systems with structured workflows, social commerce relies heavily on unstructured conversations to manage customer interactions, orders, and logistics.

  A large portion of customer interactions are repetitive, including inquiries about product availability, pricing, variations, and delivery details. During peak periods such as promotions or livestream sales, sellers often struggle to respond to the high volume of incoming messages. This leads to delayed replies, missed orders, and reduced customer satisfaction.

  In addition, inventory and order management are often handled manually through chat, making them prone to errors. Sellers may accidentally oversell products, misrecord orders, or lose track of customer requests due to the fast-paced and fragmented nature of conversations. Over time, continuous manual handling results in fatigue, increasing the likelihood of mistakes.

  Existing chatbot solutions are insufficient for this environment. Most are rule-based, stateless, and unable to understand informal language, abbreviations, or mixed-language inputs commonly used in chat. They also lack the ability to execute multi-step workflows such as extracting order details, validating stock, and coordinating shipping within a single conversation. As a result, sellers remain burdened by repetitive tasks, inefficient processes, and operational errors.

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- **Target Domain:** An AI-powered Social Commerce Operations Assistant designed to automate customer service, order processing, and logistics coordination within chat-based selling platforms. The system focuses on enabling end-to-end intelligent automation, including:
    - Understanding customer inquiries (including informal language, shortforms, and mixed-language input)
    - Extracting and structuring order details from chat conversations
    - Real-time order validation and processing
    - Inventory tracking and stock management
    - Courier comparison and shipping optimization
    - Analyzes sales trend and inventory level to generate sales report and predict stock risk
- **Proposed Solution Summary:** We propose SellerMate, an AI-powered Social Commerce Service Agent driven by advanced workflow orchestration. SellerMate transforms traditional chat-based selling by acting as an intelligent assistant that goes beyond basic chatbot functionality, enabling multi-step reasoning and seamless backend integration. When a customer interacts with the system, SellerMate will:
    - Analyze and interpret chat messages using natural language understanding (including slang and abbreviations)
    - Extract structured order information from unstructured conversations (e.g., product, quantity, variation)
    - Retrieve real-time data from inventory and order databases
    - Validate stock availability before confirming orders
    - Assist sellers in organizing and summarizing incoming orders during high-volume chats
    - Compare courier services based on cost, speed, and availability
    - Recommend optimal shipping options for seller approval
    - Generate tracking details and automatically update customers

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In addition, the system performs intelligent decision-making workflows, ensuring smooth coordination across customer communication, inventory management, and logistics. Key features include:

- Stateful conversation memory for maintaining context across chats
- Automated workflow orchestration from inquiry → order → shipment
- Real-time synchronization with inventory and order systems
- Seller dashboard for monitoring orders, stock levels, and operations

This solution significantly reduces manual workload, minimizes human error, and improves response time. At the same time, it enhances the overall customer experience by providing faster, more accurate, and reliable interactions—enabling scalable and efficient social commerce operations.

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2. Background & Business Objective

- **Background of the Problem:** With the rapid rise of social commerce platforms such as Instagram and TikTok, many small and medium-sized sellers are increasingly relying on chat-based interactions to conduct their business. Unlike traditional e-commerce platforms, social commerce lacks structured systems for order processing and inventory management, forcing sellers to manually handle operations through conversations.

  From a buyer's perspective, this creates several challenges. Sellers are often unable to respond promptly due to the overwhelming volume of repetitive inquiries. Continuous manual handling leads to fatigue, increasing the likelihood of errors such as incorrect order recording, missed messages, or delayed replies. In fast-moving scenarios such as livestream sales or comment-based purchasing (e.g., first come first serve based), sellers may struggle to accurately track orders and stock availability in real time.

  Furthermore, inventory management is often disconnected from customer interactions. This can result in situations where buyers believe they have successfully placed an order, only to be informed later that the product is out of stock, leading to frustration and dissatisfaction.

  Existing chatbot solutions are not well-suited for this environment. Most lack contextual understanding, cannot interpret informal or mixed-language inputs, and are unable to perform multi-step workflows such as extracting order details, validating stock, and coordinating delivery. As a result, sellers continue to face inefficiencies and operational challenges despite using basic automation tools.

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- **Importance of Solving This Issue:** Addressing these challenges is essential to improving both seller efficiency and buyer satisfaction in social commerce environments. By reducing reliance on repetitive manual tasks, sellers can minimize errors caused by fatigue and high workload, leading to more accurate order handling and smoother operations.

  From the buyer's perspective, faster response times and clearer communication significantly enhance the purchasing experience. Customers are more likely to trust sellers who can provide timely and accurate responses, especially in fast-paced purchasing scenarios.

  At the same time, automating key processes such as order extraction, stock validation, and shipping coordination helps prevent common issues like overselling, missed orders, and delayed fulfillment.

  Ultimately, solving this problem enables a more scalable and reliable social commerce ecosystem, where sellers can handle higher volumes of interactions efficiently while maintaining service quality, and buyers can enjoy a smoother and more trustworthy purchasing experience.

- **Strategic Fit / Impact:** This solution aligns with the growing trend of AI-driven automation in social commerce, where intelligent systems enhance human workflows rather than replace them. By leveraging advanced AI capabilities such as natural language understanding, contextual memory, and workflow orchestration, the proposed system transforms unstructured chat-based selling into a more structured and intelligent process.

  It also fits well within modern AI and cloud ecosystems, where AI models can be integrated with real-time databases, APIs, and backend services to enable end-to-end automation—from interpreting customer messages to executing operational decisions such as stock validation and shipping coordination. The key impacts include:

  - Enhancing operational efficiency for social commerce sellers

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- Reducing human error in order tracking and inventory management
- Improving customer satisfaction through faster and more accurate responses
- Enabling scalable business operations without proportional increases in workload

Overall, this positions the solution as a step toward AI-assisted social commerce, where intelligent agents support real-time decision-making and execution within conversational platforms.

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3. Product Purpose

3.1. Main Goal of the System: The primary objective of SellerMate is to provide an AI-native e-commerce operations engine that bridges the gap between front-end customer communication and back-end business management. By leveraging Z.AI GLM and Llama 3.3 70B Versatile (served via Groq), the system aims to transform repetitive, unstructured chat interactions into structured, actionable business data. Its core mission is to automate the end-to-end fulfillment lifecycle, spanning from intelligent inquiry handling and real-time inventory validation to automated courier coordination. SellerMate empowers merchants to scale their operations efficiently, ensuring that complex manual tasks are replaced by seamless, AI-driven decision-making workflows.

3.2. Intended Users (Target Audience):

- **Social Commerce Merchants (SMEs):** Small to medium-sized business owners who need to maintain high responsiveness and operational accuracy across multiple chat platforms without significantly increasing manpower.
- **Operations and Logistics Managers:** Professionals responsible for overseeing complex order fulfillment, stock management, and courier performance. They require a centralized AI assistant to minimize human error and optimize delivery costs.
- **Early-Stage Digital Entrepreneurs:** Founders looking to build a lean, automated operational infrastructure from the outset, allowing them to focus on growth and product strategy rather than routine customer service and data entry.
- **Social Commerce Customers (End-Users):** Individual shoppers who use the AI chat interface to explore products, check stock, and make purchases. While they are not involved in system operations, their experience and satisfaction play a key role in determining the system's success.

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4. System Functionalities

4.1. Description: SellerMate is an AI-native processing engine that bridges natural language chats with e-commerce workflows. It converts unstructured messages into structured database commands. By tracking the sales context continuously, it automates inventory synchronization, order validation, and logistics coordination, completely eliminating manual data entry for merchants.

4.2. Key Functionalities:

- **Stateful Context Management:** The system remembers ongoing conversations to preserve customer preferences. This allows the AI to provide coherent responses without forcing customers to repeat information.
- **Automated Workflow Orchestration:** The AI manages the entire fulfillment process from product inquiry to shipment. It automatically checks stock levels and updates order statuses. It also generates restocking recommendations that require explicit seller approval. Furthermore, it dynamically compares courier prices based on the customer location, it then calculates the final shipping fee by applying a predefined merchant profit margin before quoting the total price to the customer.
- **Intelligent Data Clarification:** When encountering slang, shortforms, or missing data, the system proactively asks targeted follow-up questions to guarantee order accuracy before proceeding.
- **Centralized Operations Dashboard:** A dedicated interface allows merchants to monitor AI interactions, review stock alerts, approve shipping recommendations, and manually intervene in complex customer service cases.

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4.3. AI Model & Prompt Design

4.3.1. Model Selection

The system utilizes a hybrid LLM architecture consisting of Z.AI GLM and Llama 3.3 70B Versatile (served via Groq), which together act as the core reasoning and language processing engines for SellerMate.

The inclusion of Z.AI GLM is primarily driven by its strong capability in context-aware, multi-turn conversation handling, allowing the system to maintain continuity across fragmented customer inputs (e.g., “PM” followed by product details in later messages). It is also highly effective in extracting structured data from unstructured chat, which is essential for transforming conversational inputs into actionable information such as product details, quantities, and delivery locations.

Complementing this, Llama 3.3 70B Versatile is leveraged for its high-performance reasoning and low-latency inference, enabling faster response times and efficient handling of complex, multi-step workflows. This makes it particularly suitable for real-time applications such as customer interaction and order processing, where speed and accuracy are critical.

Together, these models enable the system to perform end-to-end intelligent workflows, including inquiry handling, order extraction, stock validation, and courier recommendation. Both models also demonstrate strong adaptability to informal language, abbreviations, and multilingual inputs such as English, Malay, and Chinese, making them well-suited for the dynamic and unstructured nature of social commerce environments.

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4.3.2. Prompting Strategy

SellerMate adopts a multi-step agentic prompting strategy, combined with few-shot prompting for critical tasks such as order extraction and intent classification. Instead of relying on a single prompt, SellerMate decomposes complex interactions into modular steps, including intent detection (e.g., inquiry, order placement), entity extraction (e.g., product quantity, variation and location) and action execution such as database query, stock update and courier selection.

Few-shot examples are embedded within prompts to guide the model in handling domain-specific patterns, such as shorthand inputs ("1x black M", "PM price") and mixed-language phrases. This improves consistency and reduces ambiguity in structured output generation.

4.3.3. Context & Input Handling

The system we developed, SellerMate, does not enforce a strict input size limit on the client side. Instead, input constraints are primarily determined by the context windows and API limitations of both Z.AI GLM and Llama 3.3 70B Versatile. To ensure stable performance and consistent response times, each transaction is capped at a maximum of **1024 tokens**.

The system does not perform client-side truncation or chunking. All user inputs are sent to the AI model in their original form. If the input exceeds the model's allowable size, handling is deferred to the API level. When limits are exceeded, the system applies the following mechanisms:

- **API-Level Rejection:**
  Oversized inputs may be rejected by the AI server (e.g., "prompt too long"), preventing further processing.
- **Timeout Handling:**
  A timeout threshold of 40 seconds is enforced. If the model takes too long to respond (often due to large inputs), the request is automatically terminated.

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- **Graceful Failure Management:**
  The system uses error-handling mechanisms (e.g., try-catch blocks) to prevent crashes and ensure stability.
- **Error Logging & Diagnostics:**
  Detailed logs are generated, including HTTP status codes, request duration, and server error messages. This supports debugging and system monitoring.
- **Fallback Handling:**
  In cases of failure (e.g., rejection or timeout), the system safely terminates the request and can provide diagnostic feedback or trigger alternative handling (e.g., retry or escalation).

4.3.4. Fallback & Failure Behavior

Our system implements multiple fallback strategies to ensure stability and maintain a smooth user experience when AI-related failures occur. Below are the few mechanisms:

1. **Retry Mechanism**
   The system does not implement automatic retry logic. If both Z.AI and Llama 3.3 70B Versatile service fails (e.g., timeout or API error), the request is immediately handled by the fallback mechanism without additional retries. The system will alert the seller to reply to customers' messages manually. This design avoids prolonged delays and ensures faster response handling.
2. **Graceful Error Handling**
   To prevent system crashes or broken user interfaces, predefined fallback messages are used when AI responses cannot be generated.
   For example:
   - "The briefing generation failed."
   - "Failed to fetch AI explanations."
   - "AI Analyst request failed."

   These messages ensure the system remains functional and user-friendly, even during network or service disruptions.

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3. **Human-in-the-Loop (HITL) Escalation**

   The chatbot incorporates a Human-in-the-Loop (HITL) mechanism, allowing seamless transition from AI to human support:

   - **Conversation State Tracking:** All incoming messages are logged with a status such as *"unanswered"*, ensuring visibility and proper tracking.
   - **Owner Override:** Sellers can manually respond through a dedicated interface, overriding the AI when necessary.
   - **Intent-Based Escalation:** For complex queries (e.g., refunds), the AI automatically escalates the conversation and informs the customer.
   - **User-Initiated Escalation:** Customers can request human assistance (e.g., "Request owner reply"), which triggers a notification for the seller to intervene.

4. **Reliability Assurance**

   Through the combination of fallback messaging and human escalation, the system ensures:

   - Continuous operation despite AI failures
   - Reduced impact of incorrect or unusable AI responses
   - A consistent and trustworthy user experience

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5. User Stories & Use Cases

- **User Stories:**
  1. As a seller, I want the AI agent to automatically access and update my inventory database in real time when a sale is made, so that I no longer need to manage stock manually.
  2. As a seller, I want the AI agent to automatically respond to customer inquiries, so that I no longer need to reply to each message manually.
  3. As a seller, I want the AI agent to understand short forms and local slang in customers' messages (e.g., "yang ni stok ada x", " brp ni lps discount").
  4. As a seller, I want the AI agent to analyze sales trends and predict stock risk (e.g., when a certain product will out of stock)
  5. As a seller, when an item is nearly out of stock, I want the AI agent to automatically make a "Low in Stock" alert.
  6. As a seller, I want the AI agent to automatically detect the customer's location (e.g., "Tawau, Sabah") from their message and identify the most suitable courier with the best delivery price.
  7. As a seller, I want the AI agent to detect any abnormalities in product sales such as sudden massive rise or drop in sales.
  8. As a seller, I want the AI agent to generate a daily briefing about key sales performance, including total orders, revenue trends, and important operational updates.
  9. As a customer, I want the AI agent to reply to my messages/inquiry as fast as possible.

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- **Use Cases (Main Interactions):**
    1. **Real-Time Inventory Update:** When a customer places an order through chat, the AI agent automatically updates inventory if the sale is confirmed.
    2. **Automated Customer Response:** When a customer sends a message, the AI agent interprets the message and retrieves relevant information, and then automatically replies to the customer's message.
    3. **Informal Language Understanding:** The AI agent understands slang, abbreviations, and mixed-language input in customers' messages.
    4. **Stock Risk Prediction**: The AI agent analyzes and identifies products at risk of running out of stock or overstocking based on current inventory levels and sales trends.
    5. **Low Stock Alert:** The AI agent generates an alert notification to alert the seller when stock is running low.
    6. **Courier Selection Based on Location:** When a customer sends a message containing their location (state or city), the AI agent extracts location information. Then it compares available courier service and selects the most cost-effective option.
    7. **Sales Anomaly Detection:** The AI agent monitors sales data continuously and analyzes sales trends and patterns. If it detects abnormal spikes or drops, it will notify the seller.
    8. **Daily Sales Briefing**: A daily summary generated by the AI that highlights key sales performance, including total orders, revenue trends, and important operational updates for the seller.
    9. **Customer Enquiry and Place Order:** Customers can interact with the AI agent to ask about product details such as price, availability and variations.

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- **Use Case Diagram:**

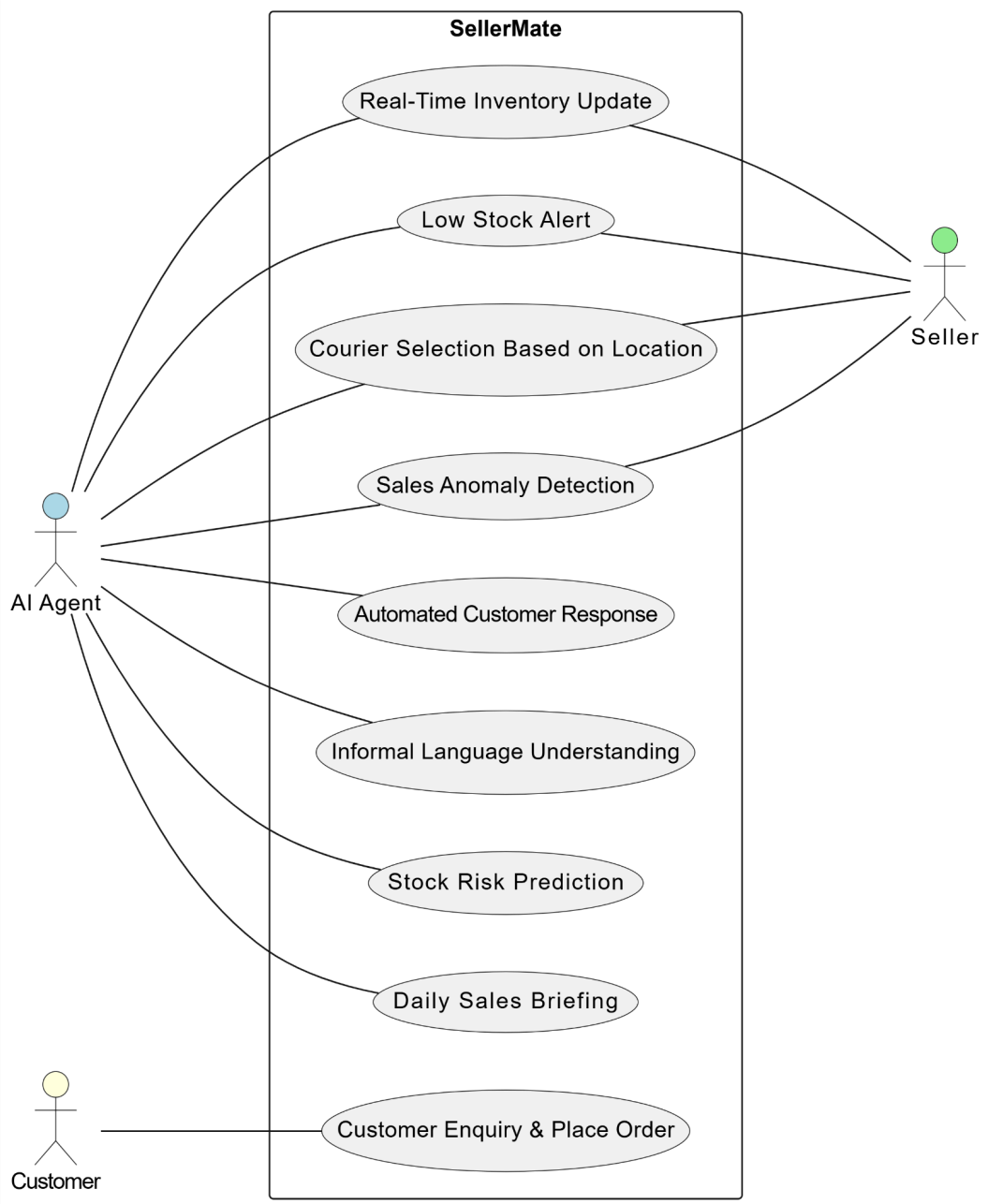

Figure 1. SellerMate Use Case Diagram

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6. Features Included (Scope Definition)

- Customers can interact with the AI agent to inquire about product details (e.g., price, availability, variations) and place orders directly through chat.
- The system automatically interprets customer messages and provides relevant, real-time responses without requiring manual seller intervention.
- The AI agent is capable of understanding slang, abbreviations, and mixed-language inputs, ensuring accurate interpretation of customer intent.
- Inventory is automatically updated when an order is confirmed, ensuring stock levels remain accurate without manual input.
- The system monitors inventory levels and generates alerts when stock falls below a predefined threshold.
- The AI extracts location information from customer messages and recommends the most suitable courier service based on cost efficiency.
- The system continuously analyzes sales data to identify unusual patterns, such as sudden spikes or drops, and notifies the seller accordingly.
- The analytical part only includes daily sales briefing and stock risk prediction which will be analyzed using current inventory level and sales trend.

7. Features Not Included (Scope Control)

- The system does not process payments or integrate with payment platforms (e.g., Stripe, FPX, e-wallets).
- Does not handle physical inventory operations such as warehousing, packing, or delivery execution.
- Complex cases like refunds, returns, and disputes require manual handling by the seller.
- Critical actions (e.g., shipping decisions, restocking) may still require seller confirmation.
- The system supports text-based chat only (no voice or speech recognition features).

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- The system is primarily designed to handle English and Malay, including informal and conversational usage, while also having the ability to understand Chinese. Other languages are supported as well; however, the system may not fully interpret or accurately understand them..
- Real-time live chat processing and analysis during high-traffic scenarios (e.g., livestream sales or flash promotions) is not included in the current scope.

8. Assumptions & Constraints

8.1 Assumptions

- **Stable Internet & API Availability**
  The system assumes reliable connectivity to external AI services such as Z.ai GLM and Llama 3.3 70B Versatile, as core functionalities depend on real-time API calls.
- **Structured Inventory Database Exists**
  It is assumed that sellers maintain a properly structured and up-to-date inventory database for accurate stock validation and order processing.
- **Customer Interaction via Text-Based Chat Only**
  The system assumes all interactions occur through text (no voice/image inputs).
- **Basic User Cooperation**
  Customers are expected to provide minimally sufficient information (e.g., product, quantity, location) for the AI to process orders effectively.

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8.2 Constraints

- **LLM Cost Constraint:** An average user session is estimated to consume approximately 300–800 tokens, with each transaction capped at 1024 tokens to control cost. To manage inference costs at scale, the system avoids redundant prompts, stores structured data instead of reprocessing conversations, and uses fallback logic to reduce unnecessary API calls.
- **Technical Constraints:** The system depends on external AI APIs (Z.AI and Groq), meaning performance is subject to third-party availability. It is limited to chat-based platforms and does not perform client-side input processing such as truncation or chunking, while language support is primarily focused on English, Malay, and Chinese.
- **Performance Constraints:** System performance is affected by AI inference latency and network conditions, with a timeout mechanism to prevent long delays. Token limits may cause input rejection, and the system is not designed to handle extremely high-volume live chat scenarios.
- **User Input:** The system relies on user prompts to function and may require clarification for incomplete or ambiguous inputs. Certain actions may require seller approval, and there remains a possibility of inaccurate AI responses that require validation or human intervention.

9. Risks & Questions Throughout Development

- **AI Response Accuracy & Hallucination**
  The AI may generate incorrect or fabricated responses when interpreting customer messages, especially in complex or ambiguous scenarios, which can affect order accuracy and user trust.

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- **Context Understanding Consistency**

  Maintaining accurate context across multi-turn conversations can be challenging, particularly when customers provide fragmented or delayed information.
- **Informal Language Interpretation**

  The use of slang, abbreviations, and mixed-language inputs may lead to misinterpretation of customer intent.
- **Order Extraction Errors**

  Incorrect extraction of key details such as product type, quantity, or variation may result in inaccurate order processing.
- **Inventory Synchronization Issues**

  Delays or mismatches between order processing and inventory updates may lead to overselling or incorrect stock levels.
- **API Reliability & Downtime**

  Dependence on external AI services such as Z.ai GLM and Llama 3.3 70B Versatile may introduce risks related to service outages, latency, or rate limits.
- **Scalability Limitations**

  System performance may degrade under higher loads as the number of simultaneous user interactions increases.
- **User Trust & Experience**

  Inconsistent, delayed, or inaccurate AI responses may negatively impact customer satisfaction and reduce trust in the system.

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